

Abd El Rahman Salah  
Supervision: Kareem Waseem

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run.do DSP48A1\_tb.v DSP48A1.v

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > DSP48A1.v

```
1 module DSP48A1 #(
2     parameter A0REG = 0,
3     parameter A1REG = 1,
4     parameter B0REG = 0,
5     parameter B1REG = 1,
6     parameter CREG = 1,
7     parameter DREG = 1,
8     parameter MREG = 1,
9     parameter PREG = 1,
10    parameter CARRYINREG = 1,
11    parameter CARRYOUTREG = 1,
12    parameter OPMODEREG = 1,
13    parameter CARRYINSEL = "OPMODE5",
14    parameter B_INPUT = "DIRECT",
15    parameter RSTTYPE = "SYNC"
16 )
17     input [17:0] A,
18     input [17:0] B,
19     input [17:0] D,
20     input [47:0] C,
21     input CLK,
22     input CARRYIN,
23     input [7:0] OPMODE,
24     input [17:0] BCIN,
25     input RSTA,
26     input RSTB,
27     input RSTM,
28     input RSTP,
29     input RSTC,
30     input RSTD,
31     input RSTCARRYIN,
32     input RSTOPMODE,
33     input CEA,
34     input CEB,
35     input CEM,
36     input CEP,
37     input CEC,
```

Ln 312, Col 10 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيًا

19:27:18 27/07/2026

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run.do DSP48A1\_tb.v DSP48A1.v X

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > DSP48A1.v

```
16 )(
38 input CED,
39 input CECARRYIN,
40 input CEOPMODE,
41 input [47:0] PCIN,
42 output [17:0] BCOUT,
43 output [47:0] PCOUT,
44 output [47:0] P,
45 output [35:0] M,
46 output CARRYOUT,
47 output CARRYOUTF
48 );
49
50 wire [17:0] b_sel;
51 assign b_sel = (B_INPUT == "CASCADE") ? BCIN : B;
52
53 reg [17:0] d_reg;
54 wire [17:0] d0;
55 generate
56 if (DREG == 1) begin : DREG_GEN
57 if (RSTTYPE == "ASYNC") begin
58 always @(posedge CLK or posedge RSTD)
59 if (RSTD) d_reg <= 18'b0;
60 else if (CED) d_reg <= D;
61 end else begin
62 always @(posedge CLK)
63 if (RSTD) d_reg <= 18'b0;
64 else if (CED) d_reg <= D;
65 end
66 assign d0 = d_reg;
67 end else begin
68 assign d0 = D;
69 end
70 endgenerate
71
72 reg [17:0] a0_reg;
73 wire [17:0] a0;
```

Ln 312, Col 10 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيًا

19:27:29 27/07/2026

File Edit Selection View Go Run Terminal Help

run.do DSP48A1\_tb.v DSP48A1.v X

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > DSP48A1.v

```
74 generate
75 if (A0REG == 1) begin : A0REG_GEN
76     if (RSTTYPE == "ASYNC") begin
77         always @(posedge CLK or posedge RSTA)
78             if (RSTA) a0_reg <= 18'b0;
79             else if (CEA) a0_reg <= A;
80     end else begin
81         always @(posedge CLK)
82             if (RSTA) a0_reg <= 18'b0;
83             else if (CEA) a0_reg <= A;
84     end
85     assign a0 = a0_reg;
86 end else begin
87     assign a0 = A;
88 end
89 endgenerate
90
91 reg [17:0] a1_reg;
92 wire [17:0] a1;
93 generate
94 if (A1REG == 1) begin : A1REG_GEN
95     if (RSTTYPE == "ASYNC") begin
96         always @(posedge CLK or posedge RSTA)
97             if (RSTA) a1_reg <= 18'b0;
98             else if (CEA) a1_reg <= a0;
99     end else begin
100         always @(posedge CLK)
101             if (RSTA) a1_reg <= 18'b0;
102             else if (CEA) a1_reg <= a0;
103     end
104     assign a1 = a1_reg;
105 end else begin
106     assign a1 = a0;
107 end
108 endgenerate
109
110 reg [17:0] b0_reg;
```

Ln 312, Col 10 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيًا

19:27:38 27/07/2026



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run.do DSP48A1\_tb.v DSP48A1.v X

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > DSP48A1.v

```

111 wire [17:0] b0;
112 generate
113 if (B0REG == 1) begin : B0REG_GEN
114     if (RSTTYPE == "ASYNC") begin
115         always @(posedge CLK or posedge RSTB)
116             if (RSTB) b0_reg <= 18'b0;
117             else if (CEB) b0_reg <= b_sel;
118     end else begin
119         always @(posedge CLK)
120             if (RSTB) b0_reg <= 18'b0;
121             else if (CEB) b0_reg <= b_sel;
122     end
123     assign b0 = b0_reg;
124 end else begin
125     assign b0 = b_sel;
126 end
127 endgenerate
128
129 reg [7:0] opmode_reg;
130 wire [7:0] opmode_int;
131 generate
132 if (OPMODEREG == 1) begin : OPMODEREG_GEN
133     if (RSTTYPE == "ASYNC") begin
134         always @(posedge CLK or posedge RSTOPMODE)
135             if (RSTOPMODE) opmode_reg <= 8'b0;
136             else if (CEOPMODE) opmode_reg <= OPMODE;
137     end else begin
138         always @(posedge CLK)
139             if (RSTOPMODE) opmode_reg <= 8'b0;
140             else if (CEOPMODE) opmode_reg <= OPMODE;
141     end
142     assign opmode_int = opmode_reg;
143 end else begin
144     assign opmode_int = OPMODE;
145 end
146 endgenerate
147

```

Ln 312, Col 10 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيا

19:28:13 27/07/2026

File Edit Selection View Go Run Terminal Help

run.do DSP48A1\_tb.v DSP48A1.v X

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > DSP48A1.v

```

148 wire [17:0] preadder_out;
149 assign preadder_out = opmode_int[4] ? (opmode_int[6] ? (d0 - b0) : (d0 + b0)) : b0;
150
151 reg [17:0] b1_reg;
152 wire [17:0] b1;
153 generate
154 if (B1REG == 1) begin : B1REG_GEN
155     if (RSTTYPE == "ASYNC") begin
156         always @(posedge CLK or posedge RSTB)
157             if (RSTB) b1_reg <= 18'b0;
158             else if (CEB) b1_reg <= preadder_out;
159     end else begin
160         always @(posedge CLK)
161             if (RSTB) b1_reg <= 18'b0;
162             else if (CEB) b1_reg <= preadder_out;
163     end
164     assign b1 = b1_reg;
165 end else begin
166     assign b1 = preadder_out;
167 end
168 endgenerate
169
170 assign BCOUT = b1;
171
172 wire signed [35:0] mult_result;
173 assign mult_result = $signed(a1) * $signed(b1);
174
175 reg [35:0] m_reg;
176 wire [35:0] m_out;
177 generate
178 if (MREG == 1) begin : MREG_GEN
179     if (RSTTYPE == "ASYNC") begin
180         always @(posedge CLK or posedge RSTM)
181             if (RSTM) m_reg <= 36'b0;
182             else if (CEM) m_reg <= mult_result;
183     end else begin
184         always @(posedge CLK)

```

Ln 312, Col 10 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيًا

19:28:22 27/07/2026

File Edit Selection View Go Run Terminal Help

run.do DSP48A1\_tb.v DSP48A1.v X

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > DSP48A1.v

```
178 if (MREG == 1) begin : MREG_GEN
183     end else begin
184         always @(posedge CLK)
185             if (RSTM) m_reg <= 36'b0;
186             else if (CEM) m_reg <= mult_result;
187     end
188     assign m_out = m_reg;
189 end else begin
190     assign m_out = mult_result;
191 end
192 endgenerate
193
194 assign M = m_out;
195
196 reg [47:0] c_reg;
197 wire [47:0] c0;
198 generate
199 if (CREG == 1) begin : CREG_GEN
200     if (RSTTYPE == "ASYNC") begin
201         always @(posedge CLK or posedge RSTC)
202             if (RSTC) c_reg <= 48'b0;
203             else if (CEC) c_reg <= C;
204     end else begin
205         always @(posedge CLK)
206             if (RSTC) c_reg <= 48'b0;
207             else if (CEC) c_reg <= C;
208     end
209     assign c0 = c_reg;
210 end else begin
211     assign c0 = C;
212 end
213 endgenerate
214
215 reg cyi_reg;
216 wire cyi;
217 generate
218 if (CARRYINREG == 1) begin : CARRYINREG_GEN
```

Ln 312, Col 10 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيًا

19:28:35 27/07/2026

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run.do DSP48A1\_tb.v DSP48A1.v X

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > DSP48A1.v

```

218 if (CARRYINREG == 1) begin : CARRYINREG_GEN
219     if (RSTTYPE == "ASYNC") begin
220         always @(posedge CLK or posedge RSTCARRYIN)
221             if (RSTCARRYIN) cyi_reg <= 1'b0;
222             else if (CECARRYIN) cyi_reg <= CARRYIN;
223     end else begin
224         always @(posedge CLK)
225             if (RSTCARRYIN) cyi_reg <= 1'b0;
226             else if (CECARRYIN) cyi_reg <= CARRYIN;
227     end
228     assign cyi = cyi_reg;
229 end else begin
230     assign cyi = CARRYIN;
231 end
232 endgenerate
233
234 wire cin_val;
235 assign cin_val = (CARRYINSEL == "CARRYIN") ? cyi :
236                 (CARRYINSEL == "OPMODES") ? opmode_int[5] : 1'b0;
237
238 wire [47:0] dab_concat;
239 assign dab_concat = {d0[11:0], a1, b1};
240
241 wire [47:0] p_out;
242
243 reg [47:0] x_mux, z_mux;
244 always @(*) begin
245     case (opmode_int[1:0])
246         2'b00: x_mux = 48'b0;
247         2'b01: x_mux = {{12{m_out[35]}}, m_out};
248         2'b10: x_mux = p_out;
249         2'b11: x_mux = dab_concat;
250         default: x_mux = 48'b0;
251     endcase
252     case (opmode_int[3:2])
253         2'b00: z_mux = 48'b0;
254         2'b01: z_mux = PCIN;

```

Ln 312, Col 10 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيًا

19:28:46 27/07/2026



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run.do DSP48A1\_tb.v DSP48A1.v

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > DSP48A1.v

```

244 always @(*) begin
252     case (opmode_int[3:2])
255         2'b10: z_mux = p_out;
256         2'b11: z_mux = c0;
257         default: z_mux = 48'b0;
258     endcase
259 end
260
261 wire signed [48:0] sum_result;
262 wire [47:0] post_result;
263 wire carry_bit;
264
265 assign sum_result = opmode_int[7] ? ({1'b0,z_mux} - ({1'b0,x_mux} + cin_val)) : ({1'b0,x_mux} + {1'b0,z_mux} + cin_val);
266 assign post_result = sum_result[47:0];
267 assign carry_bit = sum_result[48];
268
269 reg [47:0] p_reg;
270 generate
271 if (PREG == 1) begin : PREG_GEN
272     if (RSTTYPE == "ASYNC") begin
273         always @(posedge CLK or posedge RSTP)
274             if (RSTP) p_reg <= 48'b0;
275             else if (CEP) p_reg <= post_result;
276     end else begin
277         always @(posedge CLK)
278             if (RSTP) p_reg <= 48'b0;
279             else if (CEP) p_reg <= post_result;
280     end
281     assign p_out = p_reg;
282 end else begin
283     assign p_out = post_result;
284 end
285 endgenerate
286
287 assign P = p_out;
288 assign PCOUT = p_out;
289

```

Ln 312, Col 10 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيًا

19:28:56 27/07/2026

File Edit Selection View Go Run Terminal Help

run.do DSP48A1\_tb.v DSP48A1.v X

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > DSP48A1.v

```
290 reg cyo_reg;
291 wire cyo;
292 generate
293 if (CARRYOUTREG == 1) begin : CARRYOUTREG_GEN
294     if (RSTTYPE == "ASYNC") begin
295         always @(posedge CLK or posedge RSTCARRYIN)
296             if (RSTCARRYIN) cyo_reg <= 1'b0;
297             else if (CECARRYIN) cyo_reg <= carry_bit;
298     end else begin
299         always @(posedge CLK)
300             if (RSTCARRYIN) cyo_reg <= 1'b0;
301             else if (CECARRYIN) cyo_reg <= carry_bit;
302     end
303     assign cyo = cyo_reg;
304 end else begin
305     assign cyo = carry_bit;
306 end
307 endgenerate
308
309 assign CARRYOUT = cyo;
310 assign CARRYOUTF = cyo;
311
312 endmodule
```

Ln 312, Col 10 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيا

19:29:06 27/07/2026

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run.do DSP48A1\_tb.v X DSP48A1.v

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > DSP48A1\_tb.v

```
1 `timescale 1ns/1ps
2
3 module DSP48A1_tb;
4
5 reg [17:0] A, B, D;
6 reg [47:0] C;
7 reg CLK;
8 reg CARRYIN;
9 reg [7:0] OPMODE;
10 reg [17:0] BCIN;
11 reg RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRYIN, RSTOPMODE;
12 reg CEA, CEB, CEM, CEP, CEC, CED, CECARRYIN, CEOPMODE;
13 reg [47:0] PCIN;
14 wire [17:0] BCOUT;
15 wire [47:0] PCOUT;
16 wire [47:0] P;
17 wire [35:0] M;
18 wire CARRYOUT, CARRYOUTF;
19
20 integer errors;
21
22 DSP48A1 dut (
23     .A(A), .B(B), .D(D), .C(C), .CLK(CLK), .CARRYIN(CARRYIN),
24     .OPMODE(OPMODE), .BCIN(BCIN),
25     .RSTA(RSTA), .RSTB(RSTB), .RSTM(RSTM), .RSTP(RSTP), .RSTC(RSTC), .RSTD(RSTD),
26     .RSTCARRYIN(RSTCARRYIN), .RSTOPMODE(RSTOPMODE),
27     .CEA(CEA), .CEB(CEB), .CEM(CEM), .CEP(CEP), .CEC(CEC), .CED(CED),
28     .CECARRYIN(CECARRYIN), .CEOPMODE(CEOPMODE),
29     .PCIN(PCIN),
30     .BCOUT(BCOUT), .PCOUT(PCOUT), .P(P), .M(M),
31     .CARRYOUT(CARRYOUT), .CARRYOUTF(CARRYOUTF)
32 );
33
34 always #5 CLK = ~CLK;
35
36 task apply_reset;
37 begin
```

Ln 107, Col 10 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيًا

19:29:12 27/07/2026

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run.do DSP48A1\_tb.v DSP48A1.v

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > DSP48A1\_tb.v

```

37 begin
38     RSTA = 1; RSTB = 1; RSTM = 1; RSTP = 1; RSTC = 1; RSTD = 1; RSTCARRYIN = 1; RSTOPMODE = 1;
39     CEA = 0; CEB = 0; CEM = 0; CEP = 0; CEC = 0; CED = 0; CECARRYIN = 0; CEOPMODE = 0;
40     A = 0; B = 0; C = 0; D = 0; CARRYIN = 0; OPMODE = 0; BCIN = 0; PCIN = 0;
41     @(posedge CLK);
42     RSTA = 0; RSTB = 0; RSTM = 0; RSTP = 0; RSTC = 0; RSTD = 0; RSTCARRYIN = 0; RSTOPMODE = 0;
43     CEA = 1; CEB = 1; CEM = 1; CEP = 1; CEC = 1; CED = 1; CECARRYIN = 1; CEOPMODE = 1;
44 end
45 endtask
46
47 task check(input [47:0] actual, input [47:0] expected, input [127:0] name);
48 begin
49     if (actual != expected) begin
50         errors = errors + 1;
51         $display("FAIL %s expected=%0d actual=%0d time=%0t", name, expected, actual, $time);
52     end else begin
53         $display("PASS %s value=%0d time=%0t", name, actual, $time);
54     end
55 end
56 endtask
57
58 initial begin
59     CLK = 0;
60     errors = 0;
61     apply_reset;
62
63     A = 18'd5; B = 18'd3; D = 18'd10; C = 48'd0; PCIN = 48'd0; CARRYIN = 0;
64     OPMODE = 8'b00000001;
65     @(posedge CLK); @(posedge CLK); @(posedge CLK); @(posedge CLK);
66     check(P, 48'd15, "mult_5x3");
67
68     A = 18'd7; B = 18'd2; D = 18'd4;
69     OPMODE = 8'b00000001;
70     @(posedge CLK); @(posedge CLK); @(posedge CLK); @(posedge CLK);
71     check(P, 48'd14, "mult_7x2");
72
73     A = 18'd6; B = 18'd4; D = 18'd9;

```

Ln 107, Col 10 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيًا

19:29:20 27/07/2026



File Edit Selection View Go Run Terminal Help

run.do DSP48A1\_tb.v DSP48A1.v

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > DSP48A1\_tb.v

```
58 initial begin
74     OPMODE = 8'b00010001;
75     @(posedge CLK); @(posedge CLK); @(posedge CLK); @(posedge CLK);
76     check(P, 48'd78, "preadder_add_then_mult");
77
78     A = 18'd0; B = 18'd0; D = 18'd0; C = 48'd100;
79     OPMODE = 8'b00001100;
80     @(posedge CLK); @(posedge CLK); @(posedge CLK); @(posedge CLK);
81     check(P, 48'd100, "cport_passthrough");
82
83     A = 18'd8; B = 18'd5; D = 18'd0; C = 48'd200;
84     OPMODE = 8'b00001101;
85     @(posedge CLK); @(posedge CLK); @(posedge CLK); @(posedge CLK);
86     check(P, 48'd240, "multiply_add_c");
87
88     A = 18'd8; B = 18'd5; D = 18'd0; C = 48'd240;
89     OPMODE = 8'b10001101;
90     @(posedge CLK); @(posedge CLK); @(posedge CLK); @(posedge CLK);
91     check(P, 48'd200, "post_subtract");
92
93     A = 18'd0; B = 18'd0; D = 18'd0; PCIN = 48'd500;
94     OPMODE = 8'b00000100;
95     @(posedge CLK); @(posedge CLK); @(posedge CLK); @(posedge CLK);
96     check(P, 48'd500, "pcin_passthrough");
97
98     A = 18'd0; B = 18'd0; D = 18'd0; C = 48'd50;
99     OPMODE = 8'b00101100;
100    @(posedge CLK); @(posedge CLK); @(posedge CLK); @(posedge CLK);
101    check(P, 48'd51, "carryin_add");
102
103    $display("TEST COMPLETE ERRORS=%0d", errors);
104    $stop;
105 end
106
107 endmodule
```

Ln 107, Col 10 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيًا

19:29:28 27/07/2026

File Edit Selection View Go Run Terminal Help

run.do DSP48A1\_tb.v DSP48A1.v

D: > Kareem\_Waseem > Digital\_Design > Projects > Project 1 > run.do

```
1 vlib work
2 vmap work work
3 vlog DSP48A1.v
4 vlog DSP48A1_tb.v
5 vsim -voptargs=+acc work.DSP48A1_tb
6 add wave -radix decimal /DSP48A1_tb/A
7 add wave -radix decimal /DSP48A1_tb/B
8 add wave -radix decimal /DSP48A1_tb/D
9 add wave -radix decimal /DSP48A1_tb/C
10 add wave -radix binary /DSP48A1_tb/OPMODE
11 add wave -radix decimal /DSP48A1_tb/PCIN
12 add wave -radix decimal /DSP48A1_tb/M
13 add wave -radix decimal /DSP48A1_tb/P
14 add wave /DSP48A1_tb/CARRYOUT
15 add wave /DSP48A1_tb/CARRYOUTF
16 run -all
```

Ln 16, Col 9 Spaces: 4 UTF-8 CRLF [ ] Verilog Sign In

32°C مشمس جزئيًا

19:29:37 27/07/2026

Questa Sim-64 2021.1

File Edit View Compile Simulate Add Source Tools Layout Bookmarks Window Help

ColumnLayout AllColumns

Help

Layout NoDesign

Library

| Name              | Type    | Path                             |
|-------------------|---------|----------------------------------|
| mtbAvm            | Library | \$MODEL_TECH/./avm               |
| floatfixlib       | Library | \$MODEL_TECH/./floatfixlib       |
| flps_lib          | Library | \$MODEL_TECH/./flps_lib          |
| ieee              | Library | \$MODEL_TECH/./ieee              |
| ieee_env (empty)  | Library | \$MODEL_TECH/./ieee_env          |
| infact            | Library | \$MODEL_TECH/./infact            |
| mc2_lib           | Library | \$MODEL_TECH/./mc2_lib           |
| mgc_ams (empty)   | Library | \$MODEL_TECH/./mgc_ams           |
| modelsim_lib      | Library | \$MODEL_TECH/./modelsim_lib      |
| osvvm             | Library | \$MODEL_TECH/./osvvm             |
| mtbOvm            | Library | \$MODEL_TECH/./ovm-2.1.2         |
| mtbPA (empty)     | Library | \$MODEL_TECH/./pa_lib            |
| mtbRnm            | Library | \$MODEL_TECH/./rnm               |
| std               | Library | \$MODEL_TECH/./std               |
| std_developerskit | Library | \$MODEL_TECH/./std_developerskit |
| sv_std            | Library | \$MODEL_TECH/./sv_std            |
| svnoosvs          | Library | \$MODEL_TECH/./svnoosvs          |

Transcript

```

# -- Compiling module DSP48A1
#
# Top level modules:
#   DSP48A1
# End time: 19:27:00 on Jul 27,2026, Elapsed time: 0:00:00
# Errors: 0, Warnings: 0
vlog -reportprogress 300 -work work {D:/Kareem_Waseem/Digital_Design/Projects/Project 1/DSP48A1_tb.v}
# QuestaSim-64 vlog 2021.1 Compiler 2021.01 Jan 19 2021
# Start time: 19:27:00 on Jul 27,2026
# vlog -reportprogress 300 -work work D:/Kareem_Waseem/Digital_Design/Projects/Project 1/DSP48A1_tb.v
# -- Compiling module DSP48A1_tb
#
# Top level modules:
#   DSP48A1_tb
# End time: 19:27:00 on Jul 27,2026, Elapsed time: 0:00:00
# Errors: 0, Warnings: 0

```

QuestaSim>

n\_Waseem/Digital\_Design/Projects/Project 1/DSP48A1\_tb.v - Default

```

Ln#
1  `timescale 1ns/1ps
2
3  module DSP48A1_tb;
4
5      reg [17:0] A, B, D;
6      reg [47:0] C;
7      reg CLK;
8      reg CARRYIN;
9      reg [7:0] OPMODE;
10     reg [17:0] BCIN;
11     reg RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRY;
12     reg CEA, CEB, CEM, CEP, CEC, CED, CECARRYIN, CEOF;
13     reg [47:0] PCIN;
14     wire [17:0] BCOUT;
15     wire [47:0] PCOUT;
16     wire [47:0] D;

```

DSP48A1.v DSP48A1\_tb.v

32°C مشمس جزئيًا

19:30:33 27/07/2026

Questa Sim-64 2021.1

File Edit View Compile Simulate Add Transcript Tools Layout Bookmarks Window Help

100 ps

Search:

ColumnLayout AllColumns

Layout Simulate

Transcript

```
# QuestaSim-64 vlog 2021.1 Compiler 2021.01 Jan 19 2021
# Start time: 19:31:52 on Jul 27,2026
# vlog -reportprogress 300 DSP48A1.v
# -- Compiling module DSP48A1
#
# Top level modules:
#   DSP48A1
# End time: 19:31:52 on Jul 27,2026, Elapsed time: 0:00:00
# Errors: 0, Warnings: 0
# QuestaSim-64 vlog 2021.1 Compiler 2021.01 Jan 19 2021
# Start time: 19:31:52 on Jul 27,2026
# vlog -reportprogress 300 DSP48A1_tb.v
# -- Compiling module DSP48A1_tb
#
# Top level modules:
#   DSP48A1_tb
# End time: 19:31:52 on Jul 27,2026, Elapsed time: 0:00:00
# Errors: 0, Warnings: 0
# vsim -voptargs="+acc" work.DSP48A1_tb
# Start time: 19:31:52 on Jul 27,2026
# ** Note: (vsim-3812) Design is being optimized...
# Loading work.DSP48A1_tb(fast)
# Loading work.DSP48A1(fast)
# PASS      mult_5x3 value=15 time=45000
# PASS      mult_7x2 value=14 time=85000
# FAIL er_add_then_mult expected=78 actual=24 time=125000
# PASS port_passthrough value=100 time=165000
# PASS multiply_add_c value=240 time=205000
# PASS post_subtract value=200 time=245000
# PASS pcin_passthrough value=500 time=285000
# PASS      carryin_add value=51 time=325000
# TEST COMPLETE ERRORS=1
# ** Note: $stop      : DSP48A1_tb.v(104)
#      Time: 325 ns  Iteration: 1  Instance: /DSP48A1_tb
# Break in Module DSP48A1_tb at DSP48A1_tb.v line 104
VSIM 7>
```

Transcript x Wave x Dataflow x DSP48A1\_tb.v x DSP48A1.v x Processes x Objects x Library x sim x

Now: 325 ns Delta: 1 sim:/DSP48A1\_tb/#INITIAL#58

32°C مشمس جزئيًا

19:32:06 27/07/2026





Questa Sim-64 2021.1

File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help

100 ps

ColumnLayout Default

Wave - Default

| Signal              | Width       | Value    |
|---------------------|-------------|----------|
| /DSP48A1_tb/A       | 18'd0       | 0        |
| /DSP48A1_tb/B       | 18'd0       | 0        |
| /DSP48A1_tb/D       | 18'd0       | 0        |
| /DSP48A1_tb/C       | 48'd50      | 50       |
| /DSP48A1_tb/OPM...  | 8'b00101100 | 00101100 |
| /DSP48A1_tb/PCIN    | 48'd500     | 500      |
| /DSP48A1_tb/M       | 36'd0       | 0        |
| /DSP48A1_tb/P       | 48'd51      | 51       |
| /DSP48A1_tb/CARR... | 1'h0        |          |
| /DSP48A1_tb/CARR... | 1'h0        |          |

Now: 325000 ps  
Cursor 1: 0 ps

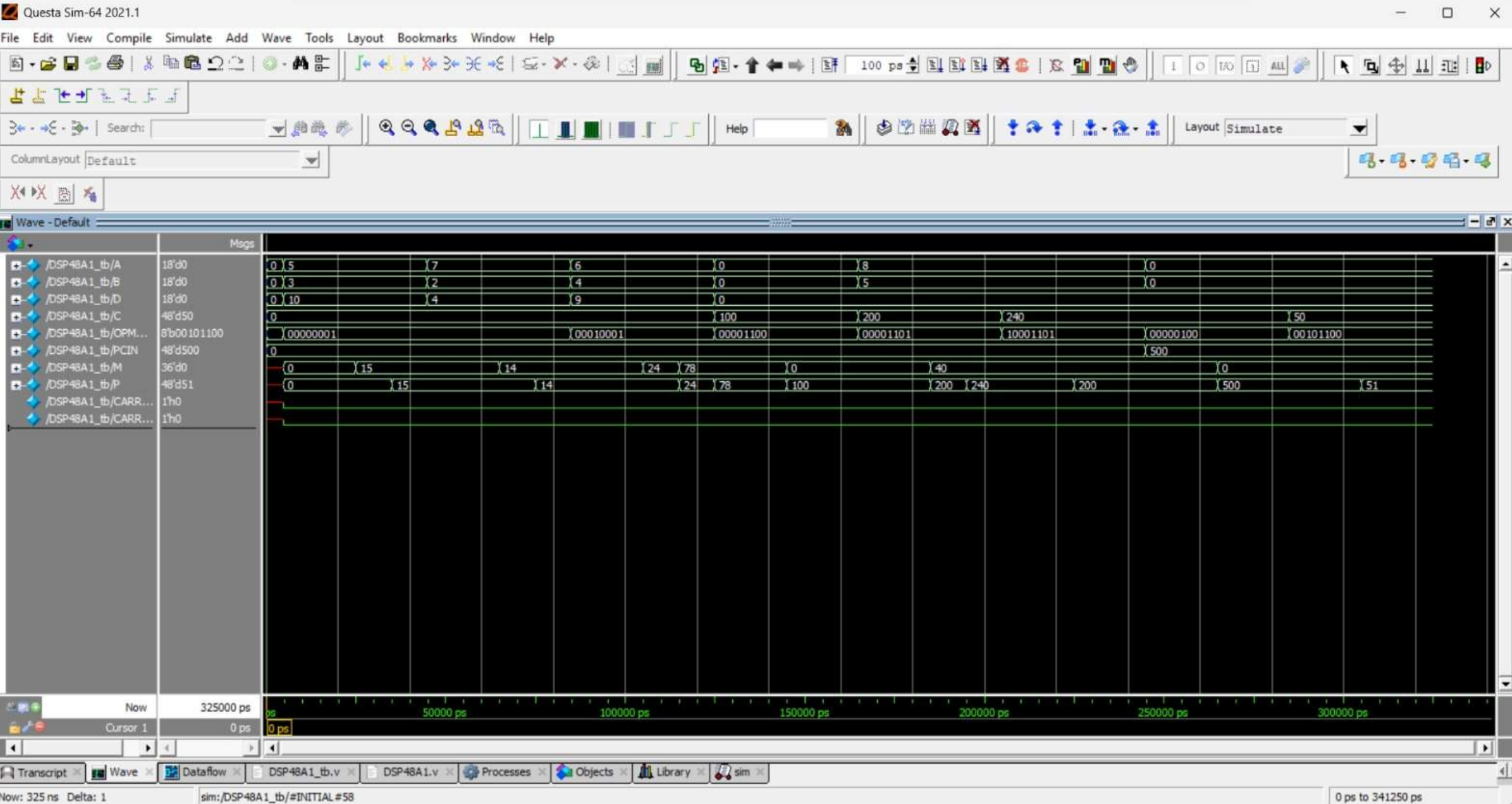
324100 ps 324200 ps 324300 ps 324400 ps 324500 ps 324600 ps 324700 ps 324800 ps 324900 ps 325000 ps

Transcript Wave Dataflow DSP48A1\_tb.v DSP48A1.v Processes Objects Library sim

Now: 325 ns Delta: 1 sim:/DSP48A1\_tb/#INITIAL #58

32°C مشمس جزئيا

19:32:23 27/07/2026



DSP481A - [D:/Kareem\_Waseem/Digital\_Design/Projects/Project 1/DSP481A/DSP481A.xpr] - Vivado 2018.2

File Edit Flow Tools Reports Window Layout View Help Quick Access

Ready

Default Layout

Flow Navigator

PROJECT MANAGER - DSP481A

Sources

Updating

Constraints (1)

Hierarchy Libraries Compile Order

Properties

Select an object to see properties

Project Summary

Settings Edit

Project name: DSP481A

Project location: D:/Kareem\_Waseem/Digital\_Design/Projects/Project 1/DSP481A

Product family: Artix-7

Project part: xc7a200tffg1156-3

Top module name: Not defined

Target language: Verilog

Simulator language: Mixed

Synthesis Implementation

Status: Not started

Messages: No errors or warnings

Part: xc7a200tffg1156-3

Tcl Console Messages Log Reports Design Runs

| Name    | Constraints | Status      | WNS | TNS | WHS | THS | TPWS | Total Power | Failed Routes | LUT | FF | BRAMs | URAM | DSP | Start | Elapsed | Run Strategy                                 |
|---------|-------------|-------------|-----|-----|-----|-----|------|-------------|---------------|-----|----|-------|------|-----|-------|---------|----------------------------------------------|
| synth_1 | constrs_1   | Not started |     |     |     |     |      |             |               |     |    |       |      |     |       |         | Vivado Synthesis Defaults (Vivado Synthesis) |
| impl_1  | constrs_1   | Not started |     |     |     |     |      |             |               |     |    |       |      |     |       |         | Vivado Implementation Defaults (Vivado Impl) |

31°C مشمس جزئيا

19:48:24 27/07/2026



DSP481A - [D:\Kareem\_Waseem\Digital\_Design\Projects\Project 1\DSP481A\DSP481A.xpr] - Vivado 2018.2

File Edit Flow Tools Reports Window Layout View Help Quick Access

Ready

Default Layout

Flow Navigator

- IP INTEGRATOR
  - Create Block Design
  - Open Block Design
  - Generate Block Design
- SIMULATION
  - Run Simulation
- RTL ANALYSIS
  - Open Elaborated Design
    - Report Methodology
    - Report DRC
    - Report Noise
    - Schematic
- SYNTHESIS
  - Run Synthesis
  - Open Synthesized Design
- IMPLEMENTATION
  - Run Implementation
  - Open Implemented Design
- PROGRAM AND DEBUG
  - Generate Bitstream

ELABORATED DESIGN - xc7a200tffg1156-3 (active)

Sources Netlist

- DSP48A1
  - Nets (820)
  - Leaf Cells (208)

Properties

Select an object to see properties

Project Summary Schematic

207 Cells 346 I/O Ports 820 Nets

Tcl Console Messages Log Reports Design Runs

| Name    | Constraints | Status      | WNS | TNS | WHS | THS | TPWS | Total Power | Failed Routes | LUT | FF | BRAMs | URAM | DSP | Start | Elapsed | Run Strategy                                 |
|---------|-------------|-------------|-----|-----|-----|-----|------|-------------|---------------|-----|----|-------|------|-----|-------|---------|----------------------------------------------|
| synth_1 | constrs_1   | Not started |     |     |     |     |      |             |               |     |    |       |      |     |       |         | Vivado Synthesis Defaults (Vivado Synthesis) |
| impl_1  | constrs_1   | Not started |     |     |     |     |      |             |               |     |    |       |      |     |       |         | Vivado Implementation Defaults (Vivado Impl) |

31°C مشمس جزئيا

19:53:08 27/07/2026

DSP481A - [D:\Kareem\_Waseem\Digital\_Design\Projects\Project 1\DSP481A\DSP481A.xpr] - Vivado 2018.2

File Edit Flow Tools Reports Window Layout View Help Quick Access

Synthesis Complete ✓

Default Layout

Flow Navigator

ELABORATED DESIGN - xc7a200tffg1156-3 (active)

Sources x Netlist

Constraints (1)

constrs\_1 (1)

DSP481A.xdc

Simulation Sources (1)

sim\_1 (1)

DSP48A1 (DSP48A1.v)

Hierarchy Libraries Compile Order

Properties

Select an object to see properties

Project Summary x Schematic

207 Cells 346 I/O Ports 820 Nets

Synthesis Completed

Synthesis successfully completed.

Next

☒ Run Implementation

☐ Open Synthesized Design

☐ View Reports

☐ Don't show this dialog again

OK Cancel

Tcl Console Messages x Log Reports Design R

Warning (64) Info (33) Status (24) Show All

General Messages (21 warnings, 8 infos)

- [Synth 8-2490] overwriting previous definition of module DSP48A1 [DSP48A1.v:1]
- [Synth 8-6157] synthesizing module 'DSP48A1' [DSP48A1.v:1]
- [Synth 8-226] default block is never used [DSP48A1.v:245] (1 more like this)
- [Synth 8-6014] Unused sequential element CARRYINREG\_GEN.cyl\_reg\_reg was removed. [DSP48A1.v:225]
- [Synth 8-6155] done synthesizing module 'DSP48A1' (1#1) [DSP48A1.v:1]
- [Synth 8-3331] design DSP48A1 has unconnected port CARRYIN (18 more like this)
- [Device 21-403] Loading part xc7a200tffg1156-3
- [Project 1-570] Preparing netlist for logic optimization

31°C مشمس جزئيا

19:56:30 27/07/2026

DSP481A - [D:/Kareem\_Waseem/Digital\_Design/Projects/Project 1/DSP481A/DSP481A.xpr] - Vivado 2018.2

File Edit Flow Tools Reports Window Layout View Help Quick Access

Synthesis Complete ✓

Default Layout

Flow Navigator

- IP INTEGRATOR
  - Create Block Design
  - Open Block Design
  - Generate Block Design
- SIMULATION
  - Run Simulation
- RTL ANALYSIS
  - Open Elaborated Design
    - Report Methodology
    - Report DRC
    - Report Noise
    - Schematic
- SYNTHESIS**
  - Run Synthesis
  - Open Synthesized Design
    - Constraints Wizard
    - Edit Timing Constraints
    - Set Up Debug
    - Report Timing Summary
    - Report Clock Networks
    - Report Clock Interaction

SYNTHESIZED DESIGN - xc7a200tfg1156-3 (active)

Sources Netlist

- DSP48A1
  - Nets (1161)
  - Leaf Cells (753)

Properties

Select an object to see properties

Project Summary Device

X0Y4 X1Y4

X0Y3 X1Y3

X0Y2 X1Y2

X0Y1 X1Y1

X0Y0 X1Y0

Tcl Console Messages Log Reports Design Runs

INFO: [Project 1-570] Preparing netlist for logic optimization

Parsing XDC File [D:/Kareem\_Waseem/Digital\_Design/Projects/Project 1/DSP481A.xdc]

Finished Parsing XDC File [D:/Kareem\_Waseem/Digital\_Design/Projects/Project 1/DSP481A.xdc]

INFO: [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).

INFO: [Project 1-111] Unisim Transformation Summary:

No Unisim elements were transformed.

Type a Tcl command here

31°C مشمس جزئيًا

27/07/2026 19:57:43

## Flow Navigator

- Report Methodology
- Report DRC
- Report Noise
- Schematic

## SYNTHESIS

- Run Synthesis
- Open Synthesized Design
  - Constraints Wizard
  - Edit Timing Constraints
  - Set Up Debug
  - Report Timing Summary
  - Report Clock Networks
  - Report Clock Interaction
  - Report Methodology
  - Report DRC
  - Report Noise
  - Report Utilization
- Report Power
- Schematic

## IMPLEMENTATION

- Run Implementation
- Open Implemented Design

## Report Timing Summary

Generate a timing summary to understand if the design met timing.

Results name:

### Options

Advanced

Timer Settings

### Report

Path delay type:

☒ Report unconstrained paths

☐ Report datasheet

### Path Limits

Maximum number of paths per clock or path group:

Maximum number of worst paths per endpoint:

### Path Display

Display paths with slack less than:  ☒ Use default (1e+30)

Significant digits:

Command: `report_timing_summary -delay_type min_max -report_unconstrained -check_timing_verbose -max_paths 10 -input_pins -routable_nets -name timing_1`

☒ Open in a new tab

☐ Open in Timing Analysis layout



Open in the specified layout

OK

Cancel

Type a Tcl command here

Report timing summary to understand if the design met timing



DSP481A - [D:/Kareem\_Waseem/Digital\_Design/Projects/Project 1/DSP481A/DSP481A.xpr] - Vivado 2018.2

File Edit Flow Tools Reports Window Layout View Help Quick Access

Synthesis Complete ✓

Default Layout

Flow Navigator

- Report Methodology
- Report DRC
- Report Noise
- Schematic
- SYNTHESIS**
  - Run Synthesis
  - Open Synthesized Design
    - Constraints Wizard
    - Edit Timing Constraints
    - Set Up Debug
    - Report Timing Summary
    - Report Clock Networks
    - Report Clock Interaction
    - Report Methodology
    - Report DRC
    - Report Noise
    - Report Utilization
    - Report Power
    - Schematic
- IMPLEMENTATION
  - Run Implementation
  - Open Implemented Design

SYNTHESIZED DESIGN - xc7a200tfg1156-3 (active)

Sources Netlist

- DSP48A1
  - Nets (1161)
  - Leaf Cells (753)

Properties

Select an object to see properties

Project Summary Device

X0Y4 X1Y4

X0Y3 X1Y3

X0Y2 X1Y2

X0Y1 X1Y1

X0Y0 X1Y0

Tcl Console Messages Log Reports Design Runs Timing

Design Timing Summary

General Information

Timer Settings

Design Timing Summary

Clock Summary (1)

- Check Timing (326)
- Intra-Clock Paths
- Inter-Clock Paths
- Other Path Groups

Timing Summary - timing\_1

| Setup                                | Hold                             | Pulse Width                                       |
|--------------------------------------|----------------------------------|---------------------------------------------------|
| Worst Negative Slack (WNS): 5.925 ns | Worst Hold Slack (WHS): 0.249 ns | Worst Pulse Width Slack (WPWS): 4.500 ns          |
| Total Negative Slack (TNS): 0.000 ns | Total Hold Slack (THS): 0.000 ns | Total Pulse Width Negative Slack (TPWS): 0.000 ns |
| Number of Failing Endpoints: 0       | Number of Failing Endpoints: 0   | Number of Failing Endpoints: 0                    |
| Total Number of Endpoints: 85        | Total Number of Endpoints: 85    | Total Number of Endpoints: 161                    |

All user specified timing constraints are met.

31°C مشمس جزئيًا

20:01:21 27/07/2026

DSP481A - [D:/Kareem\_Waseem/Digital\_Design/Projects/Project 1/DSP481A/DSP481A.xpr] - Vivado 2018.2

File Edit Flow Tools Reports Window Layout View Help Quick Access

Synthesis Complete ✓

Default Layout

Flow Navigator

- Report Methodology
- Report DRC
- Report Noise
- Schematic
- SYNTHESIS**
  - Run Synthesis
  - Open Synthesized Design
    - Constraints Wizard
    - Edit Timing Constraints
    - Set Up Debug
    - Report Timing Summary
    - Report Clock Networks
    - Report Clock Interaction
    - Report Methodology
    - Report DRC
    - Report Noise
    - Report Utilization
    - Report Power
    - Schematic
- IMPLEMENTATION
  - Run Implementation
  - Open Implemented Design

SYNTHESIZED DESIGN - xc7a200tfg1156-3 (active)

Sources Netlist

- DSP48A1
  - Nets (1161)
  - Leaf Cells (753)

Properties

Select an object to see properties

Project Summary Device

X0Y4 X1Y4

X0Y3 X1Y3

X0Y2 X1Y2

X0Y1 X1Y1

X0Y0 X1Y0

Tcl Console Messages Log Reports Design Runs Utilization x Timing

Hierarchy

| Name    | Slice LUTs (134600) | Slice Registers (269200) | DSPs (740) | Bonded IOB (500) | BUFGCTRL (32) |
|---------|---------------------|--------------------------|------------|------------------|---------------|
| DSP48A1 | 230                 | 159                      | 1          | 327              | 1             |

utilization\_1

31°C مشمس جزئيًا

20:02:18 27/07/2026

DSP481A - [D:/Kareem\_Waseem/Digital\_Design/Projects/Project 1/DSP481A/DSP481A.xpr] - Vivado 2018.2

File Edit Flow Tools Reports Window Layout View Help Quick Access

Implementation Complete ✓

Default Layout

Flow Navigator

- Report Methodology
- Report DRC
- Report Noise
- Schematic

SYNTHESIS

- Run Synthesis
- Open Synthesized Design
  - Constraints Wizard
  - Edit Timing Constraints
- Set Up Debug
- Report Timing Summary
  - Report Clock Networks
  - Report Clock Interaction
- Report Methodology
- Report DRC
- Report Noise
- Report Utilization
- Report Power
- Schematic

IMPLEMENTATION

- Run Implementation
- Open Implemented Design
  - Constraints Wizard

IMPLEMENTS DESIGN - xc7a200tffg1156-3 (active)

Sources Netlist

- DSP48A1
  - Nets (1094)
  - Leaf Cells (771)

Properties

Select an object to see properties

Project Summary Device

Timing

Design Timing Summary

General Information

Timer Settings

Design Timing Summary

Clock Summary (1)

- Check Timing (326)
- Intra-Clock Paths
- Inter-Clock Paths
- Other Path Groups

Timing Summary - impl\_1 (saved)

| Setup                                | Hold                             | Pulse Width                                       |
|--------------------------------------|----------------------------------|---------------------------------------------------|
| Worst Negative Slack (WNS): 4.538 ns | Worst Hold Slack (WHS): 0.343 ns | Worst Pulse Width Slack (WPWS): 4.500 ns          |
| Total Negative Slack (TNS): 0.000 ns | Total Hold Slack (THS): 0.000 ns | Total Pulse Width Negative Slack (TPWS): 0.000 ns |
| Number of Failing Endpoints: 0       | Number of Failing Endpoints: 0   | Number of Failing Endpoints: 0                    |
| Total Number of Endpoints: 104       | Total Number of Endpoints: 104   | Total Number of Endpoints: 180                    |

All user specified timing constraints are met.

31°C مشمس جزئيا

20:13:05 27/07/2026

DSP481A - [D:/Kareem\_Waseem/Digital\_Design/Projects/Project 1/DSP481A/DSP481A.xpr] - Vivado 2018.2

File Edit Flow Tools Reports Window Layout View Help Quick Access

Implementation Complete ✓

Default Layout

Flow Navigator

- Report Methodology
- Report DRC
- Report Noise
- Schematic

SYNTHESIS

- Run Synthesis
- Open Synthesized Design
  - Constraints Wizard
  - Edit Timing Constraints
- Set Up Debug
- Report Timing Summary
  - Report Clock Networks
  - Report Clock Interaction
- Report Methodology
- Report DRC
- Report Noise
- Report Utilization
- Report Power
- Schematic

IMPLEMENTATION

- Run Implementation
- Open Implemented Design
  - Constraints Wizard

IMPLEMENTS DESIGN - xc7a200tffg1156-3 (active)

Sources Netlist

- DSP48A1
  - Nets (1094)
  - Leaf Cells (771)

Properties

Select an object to see properties

Project Summary Device

Timing

Design Timing Summary

General Information

Timer Settings

Design Timing Summary

Clock Summary (1)

- Check Timing (326)
- Intra-Clock Paths
- Inter-Clock Paths
- Other Path Groups

Timing Summary - impl\_1 (saved)

| Setup                                | Hold                             | Pulse Width                                       |
|--------------------------------------|----------------------------------|---------------------------------------------------|
| Worst Negative Slack (WNS): 4.538 ns | Worst Hold Slack (WHS): 0.343 ns | Worst Pulse Width Slack (WPWS): 4.500 ns          |
| Total Negative Slack (TNS): 0.000 ns | Total Hold Slack (THS): 0.000 ns | Total Pulse Width Negative Slack (TPWS): 0.000 ns |
| Number of Failing Endpoints: 0       | Number of Failing Endpoints: 0   | Number of Failing Endpoints: 0                    |
| Total Number of Endpoints: 104       | Total Number of Endpoints: 104   | Total Number of Endpoints: 180                    |

All user specified timing constraints are met.

31°C مشمس جزئيا

20:17:15 27/07/2026



DSP481A - [D:/Kareem\_Waseem/Digital\_Design/Projects/Project 1/DSP481A/DSP481A.xpr] - Vivado 2018.2

File Edit Flow Tools Reports Window Layout View Help Quick Access

Implementation Complete ✓

Default Layout

Flow Navigator

- Report Methodology
- Report DRC
- Report Noise
- Schematic
- SYNTHESIS
  - Run Synthesis
  - Open Synthesized Design
    - Constraints Wizard
    - Edit Timing Constraints
  - Set Up Debug
  - Report Timing Summary
  - Report Clock Networks
  - Report Clock Interaction
  - Report Methodology
  - Report DRC
  - Report Noise
  - Report Utilization
  - Report Power
  - Schematic
- IMPLEMENTATION
  - Run Implementation
  - Open Implemented Design
    - Constraints Wizard

IMPLEMENTED DESIGN - xc7a200tfg1156-3 (active)

Sources Netlist

- DSP48A1
  - Nets (1094)
  - Leaf Cells (771)

Properties

Select an object to see properties

Project Summary Device

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Settings

Summary (0.278 W, Margin: N/A)

Power Supply

Utilization Details

- Hierarchical (0.147 W)
- Clocks (0.002 W)
- Signals (0.01 W)
  - Data (0.01 W)
  - Clock Enable (<0.001 W)
  - Set/Reset (0 W)
- Logic (0.002 W)
- DSP (0.001 W)

impl\_1 (saved)

On-Chip Power

Dynamic: 0.147 W (53%)

Device Static: 0.131 W (47%)

Power Breakdown:

- Clocks: 0.002 W (1%)
- Signals: 0.010 W (7%)
- Logic: 0.002 W (1%)
- DSP: 0.001 W (1%)
- I/O: 0.132 W (90%)

Summary Data:

| Parameter                           | Value           |
|-------------------------------------|-----------------|
| Total On-Chip Power:                | 0.278 W         |
| Design Power Budget:                | Not Specified   |
| Power Budget Margin:                | N/A             |
| Junction Temperature:               | 25.4°C          |
| Thermal Margin:                     | 74.6°C (50.8 W) |
| Effective SJA:                      | 1.5°C/W         |
| Power supplied to off-chip devices: | 0 W             |
| Confidence level:                   | Low             |

31°C مشمس جزئيا

20:18:26 27/07/2026